CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 10: **Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Cardiac muscles never get fatigued. Reason (R): They are voluntary muscles. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The number of annual rings in a tree trunk indicates: 1

(a) its height **(b)** its age **(c)** its species **(d)** its weight

3. The tissue of the cornea of the eye is largely: 1

(a) fat
(b) transparent epithelium and connective tissue
(c) bone
(d) muscle

4. Which of the following is essential for the conduction of impulses along a nerve fibre? 1

(a) sclerenchyma **(b)** xylem
(c) keratin **(d)** electrical and chemical changes

5. Red ink colours the stem of a balsam plant because: 1

(a) epidermis absorbs **(b)** cork absorbs
(c) phloem carries water **(d)** xylem carries water

6. Assertion (A): Aquatic plants have well-developed aerenchyma. Reason (R): Air spaces provide buoyancy and store gases. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A wide annual ring indicates a year with: 1

(a) drought **(b)** no growth
(c) favourable growth conditions **(d)** frost

8. The respiratory tract is lined with cilia to: 1

(a) carry oxygen **(b)** absorb food
(c) contract **(d)** sweep dust and mucus

9. A muscle that contracts and relaxes rhythmically for a lifetime without fatigue is: 1

(a) skeletal muscle **(b)** areolar tissue **(c)** smooth muscle **(d)** cardiac muscle

10. The cuticle on the leaf epidermis is made of: 1

(a) lignin **(b)** cellulose **(c)** protein **(d)** wax-like cutin

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do muscles need a rich blood supply? 2

12. Why do the muscle cells have many mitochondria compared with an epithelial cell? 2

13. How is the nervous tissue different from the muscular tissue in function? 2

14. Explain the importance of cartilage at the ends of long bones. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain, with examples, how the different tissues of the leaf cooperate in photosynthesis. 3

16. Explain why an organ transplant requires matching tissues. 3

17. A man has a damaged spinal cord. Explain which tissue is affected and the consequences. 3

OR

Explain how joints and muscles together produce the movement of a limb with the example of the elbow. 3

18. Explain the use of stem cells to repair damaged tissue. 3

OR

Compare the roles of the cell, tissue, organ and organ system in a living organism with an example from the digestive system. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the types of joints and explain the structure of a synovial joint with the tissues present and what happens in arthritis. 5

OR

Analyse the statement 'Division of labour among cells leads to complexity in living organisms' using the levels of organisation and examples of specialised tissues in plants and animals. 5

20. A student cuts a transverse section of a root and a stem. Compare the arrangement of tissues in both with the functions: epidermis, cortex, endodermis, pith, vascular bundles. 5

OR

Explain how the tissues of the skin, intestine, lung and blood vessels are all lined by epithelium but differ in structure, and relate to the functions with a table. 5

END OF QUESTION PAPER — **Total: 40 Marks**